

# LAB EXPERIMENT RECORD

## Set 15: Product Sales Analysis

|                           |                                           |
|---------------------------|-------------------------------------------|
| Department                | CSE-DS                                    |
| Course Code / Course Name | DSA0601 / Data Handling and Visualization |
| Experiment                | Data Visualization using R                |
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| Register Number           | 192473018                                 |
| Date                      | 08 October 2026                           |
| Day                       | Thursday                                  |

### Dataset: Monthly Product Sales

| Product ID | Product Name | January Sales | February Sales | March Sales |
|------------|--------------|---------------|----------------|-------------|
| 1          | Product A    | 2000          | 2200           | 2400        |
| 2          | Product B    | 1500          | 1800           | 1600        |
| 3          | Product C    | 1200          | 1400           | 1100        |

### 1. Grouped Bar Chart

#### Aim

To create a grouped bar chart to visualize the sales of each product during the first quarter.

#### R Code

R

```
sales <- matrix(c(
  2000, 2200, 2400,
  1500, 1800, 1600,
  1200, 1400, 1100
```

```
), nrow = 3, byrow = TRUE)

colnames(sales) <- c("January", "February", "March")

rownames(sales) <- c("Product A", "Product B", "Product C")

barplot(sales,

  beside = TRUE,

  main = "First Quarter Product Sales",

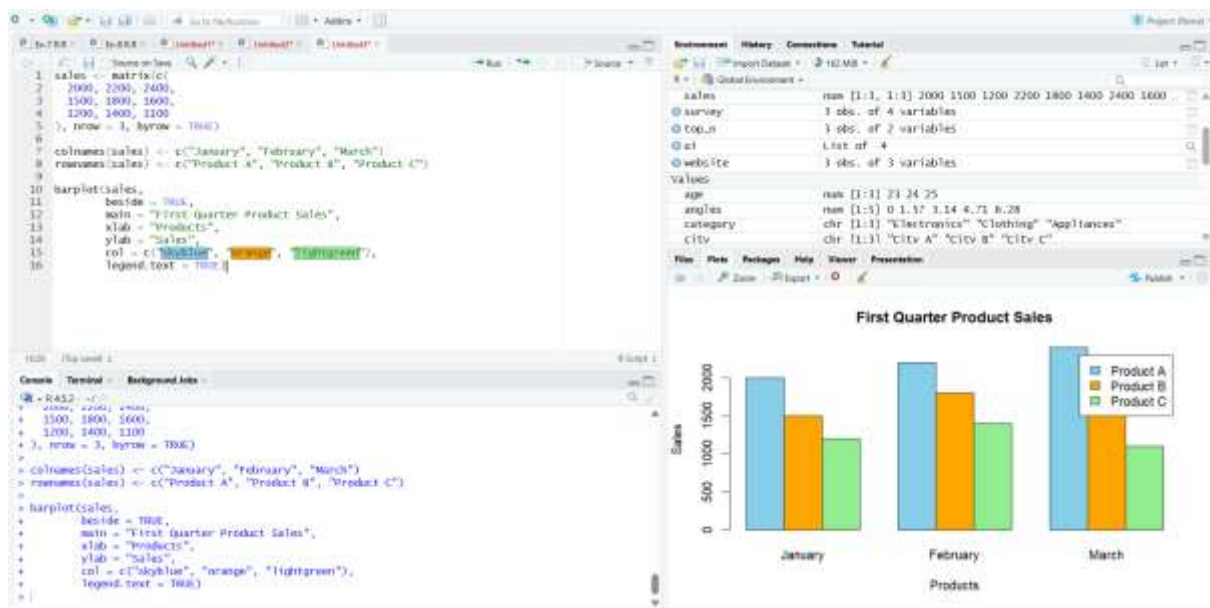
  xlab = "Products",

  ylab = "Sales",

  col = c("skyblue", "orange", "lightgreen"),

  legend.text = TRUE)
```

## Output



A grouped bar chart showing January, February and March sales for Product A, Product B and Product C.

## Observation

Product A has the highest sales in all three months. Product C has the lowest sales in March.

## 2. Stacked Area Chart

### Aim

To create a stacked area chart representing the overall sales trend of all products over the three months.

### **R Code**

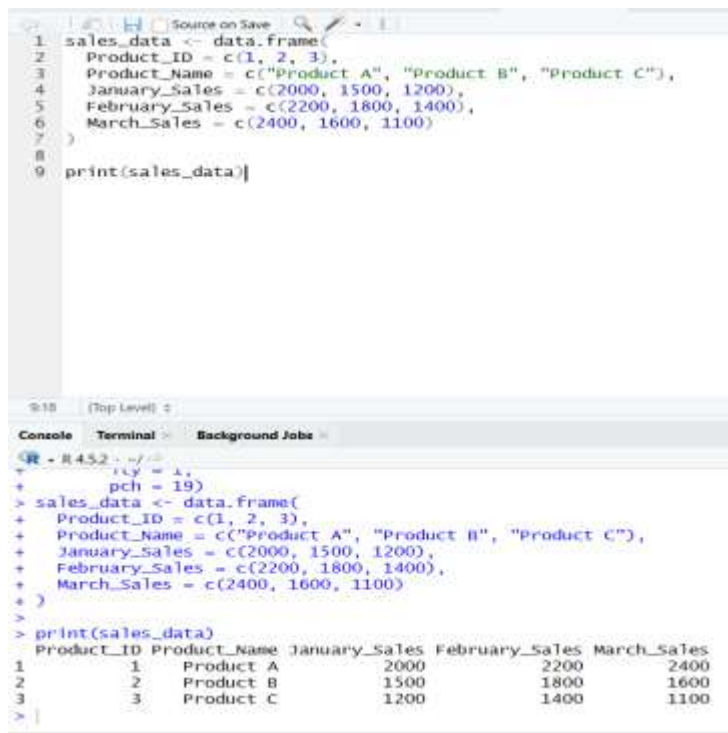
R

```
sales <- matrix(c(
  2000, 2200, 2400,
  1500, 1800, 1600,
  1200, 1400, 1100
), nrow = 3, byrow = TRUE)
colnames(sales) <- c("January", "February", "March")
months <- 1:3
matplot(months,
  t(sales),
  type = "o",
  pch = 19,
  xlab = "Months",
  ylab = "Sales",
  main = "Overall Monthly Sales Trend",
  xaxt = "n")
axis(1, at = 1:3, labels = c("January", "February", "March"))
legend("topleft",
  legend = c("Product A", "Product B", "Product C"),
  lty = 1,
  pch = 19)
```

### **Output**



## Output



```
1 sales_data <- data.frame(  
2   Product_ID = c(1, 2, 3),  
3   Product_Name = c("Product A", "Product B", "Product C"),  
4   January_Sales = c(2000, 1500, 1200),  
5   February_Sales = c(2200, 1800, 1400),  
6   March_Sales = c(2400, 1600, 1100)  
7 )  
8  
9 print(sales_data)
```

R - R4.5.2 - ~/...  
+ rpy = 1,  
+ pch = 19)  
> sales\_data <- data.frame(  
+ Product\_ID = c(1, 2, 3),  
+ Product\_Name = c("Product A", "Product B", "Product C"),  
+ January\_Sales = c(2000, 1500, 1200),  
+ February\_Sales = c(2200, 1800, 1400),  
+ March\_Sales = c(2400, 1600, 1100)  
+ )  
> print(sales\_data)

|   | Product_ID | Product_Name | January_Sales | February_Sales | March_Sales |
|---|------------|--------------|---------------|----------------|-------------|
| 1 | 1          | Product A    | 2000          | 2200           | 2400        |
| 2 | 2          | Product B    | 1500          | 1800           | 1600        |
| 3 | 3          | Product C    | 1200          | 1400           | 1100        |

## Observation

The table clearly displays the monthly sales of each product for the first quarter.

## 4. Tableau Dashboard

### Aim

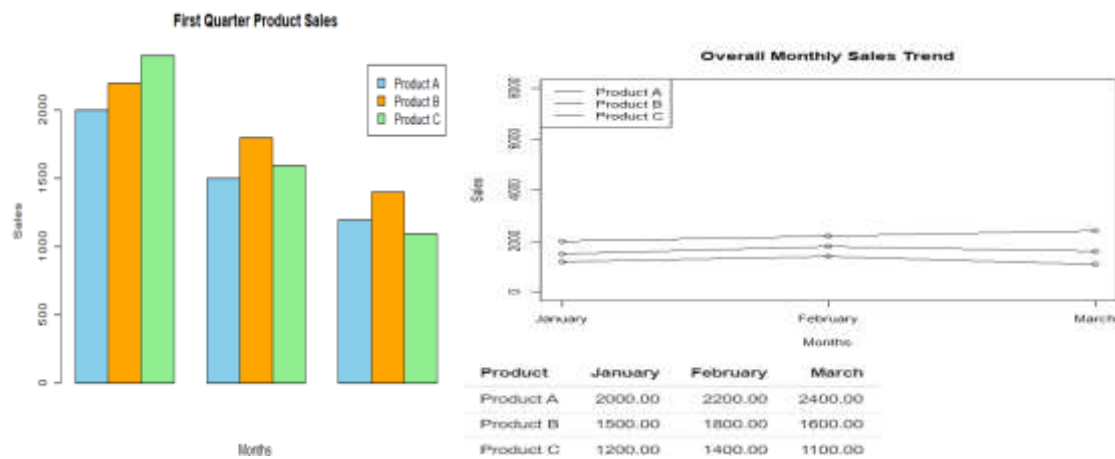
To develop an interactive dashboard combining the grouped bar chart, stacked area chart and monthly sales table.

### Tableau Procedure

1. Open **Tableau**.
2. Import the **Monthly Product Sales** dataset.
3. Create a **Grouped Bar Chart** using Product Name and monthly sales.
4. Create a **Stacked Area Chart** using Month and Sales.
5. Create a **Table** containing Product ID, Product Name, January Sales, February Sales and March Sales.
6. Select **Dashboard** → **New Dashboard**.
7. Drag the grouped bar chart into the dashboard.
8. Drag the stacked area chart into the dashboard.

9. Drag the sales table into the dashboard.
10. Add suitable titles and labels.
11. Arrange the charts and table neatly.
12. Save the dashboard.

### Output:



### Result

The monthly product sales data was successfully analyzed using a grouped bar chart, stacked area chart and table. The Tableau dashboard provides an interactive view of product sales during the first quarter.